



Impact of E-Banking Service Factors on Client Satisfaction and Sustainable Loyalty in the Republic of Kosovo: A Case Study

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Abstract: The process of digital transformation entails the development of inclusive and reliable financial infrastructure considered to be crucial for economic stability, especially in developing and transition economies. The financial sector of Kosovo is mainly dominated by commercial banks which heavily rely on the deposits of private clients as the main source of funding, thus customer loyalty is essential for their funding stability. In line with the Sustainable Development Goal 9 which underlies the significance of innovation and sustainable industrialization, the purpose of this research is to investigate the dimensions of e-banking service quality, service price, and the impact of socio-demographic factors on customer satisfaction and loyalty within the banking sector in the Republic of Kosovo. This study applied both qualitative and quantitative methods to collect data and analyze research results. It was noted that the service quality of e-banking emerged with a positive impact on customer satisfaction, yet exerted no significantly direct impact on customer loyalty. Reliability, as a dimension of e-banking service quality, turned out to be the key factor that positively influenced customer satisfaction, followed by responsiveness and sensitivity. Besides, financial innovation was positively perceived by the bank clients in developing countries, despite the adverse impact derived from the negative relationship between price and loyalty. Therefore, precisely identifying and perceiving potential factors that influence e-banking satisfaction and client loyalty is crucial to support efforts striving towards financial inclusion. The findings in the current study suggest that a balanced approach that encourages innovation while maintaining fair pricing strategies is indispensable to ensuring that the positive impact of e-banking translates into a bank's financial stability in the developing countries. This study offers insightful knowledge for commercial banks and regulators who are interested in factors affecting progressive digital financial inclusion in emerging banking sectors.

Keywords: Electronic banking; Customer satisfaction; Customer loyalty; Modified e-SERVQUAL

1. Introduction

One of the Sustainable Development Goals (SDGs) advocated by the United Nations, SDG 9 “Industries, Innovation and Infrastructure”, highlights that investments in resilient infrastructure, sustainable industrial development, and technological innovation significantly drive economic growth and social development by providing people with access to opportunities (United Nations, 2015). In light of this, the banking sector presents a fundamental function in sustainable development facilitated by their financial intermediation function (Vo et al., 2021). People keep money in banks for security and convenient transactional purposes, rendering trust in the banking sector crucial for the stability of a country's financial system. It is apparently evident that, banks should employ emerging methods to retain customers and achieve planned business results in order to boost their competitiveness (Karjaluoto et al., 2002). One of the most important concepts in service marketing is service loyalty, as loyal customers are fundamental to lasting and successful business results (Caruana, 2002).

E-banking or online banking as a financial innovation could enable customers' effortless access to their accounts anytime and anywhere; it offers banks less geographical restrictions for their distribution of banking services. Nevertheless, despite the findings that acceptance of online banking was associated with lower costs and improved profitability, it has not yet been considered a substitute of physical banks but rather a complementary distribution

channel (DeYoung et al., 2007; Hernando & Nieto, 2007). This finding is in line with the relatively stable number of bank branches and sub-branches in the banking sector of the Republic of Kosovo during the last five years although the number of e-banking accounts was more than doubled during the same period.

In a highly competitive market where services could be easily replicated, identifying the factors that influence customer loyalty remains essential for banks. So many researchers have shown interest in exploring service quality in their works (Buttle, 1996; Parasuraman et al., 1985; Zeithaml, 1988). Given that customer feedback is generally not included in the course of service development (Vermeulen, 2005), there is an argument that pricing strategies are frequently formulated as a response to the competitors' pricing actions (Streiter et al., 1997). There is evidence that customers generally have difficulties in assessing similar financial services and a significant number are unable to make informed financial decisions as well (Financial Services Authority, 2006). The question arises as to what factors determine the perception of service quality and customer satisfaction with financial services. While the study of Stafford (1996) continues to serve as a reference in numerous research, it provides insights into how the perceptions of service quality vary in different demographic groups, leading to implications for the pricing strategies.

This empirical research investigates the impact of factors and the dimensions of e-banking service quality on client satisfaction and their loyalty to banks operating in the Republic of Kosovo. It aims to contribute to a broader literature on digital banking services by extending the e-SERVQUAL service quality model to assess client satisfaction and loyalty in Kosovo, a developing economy with a rapidly digitalizing banking sector and a limited body of research on the factors influencing this relationship and client behavior. By combining banking service quality with pricing and demographic factors into a single framework, the analysis offers a new perspective of how clients perceive digital banking services and how their loyalty develops within a particular socioeconomic context, accounting for price and quality implications as well as demographic variables.

2. Review of Literature

2.1 Service of E-Banking

Rapid changes in the external environment such as regulation modifications, increased competitions, and swift technological changes, have redefined the offering of banking services (John & Harborne, 2003). The pace of technology has revolutionized communication, developed industrial infrastructure, and created internet services to replace traditional banking services (Rafinejad, 2007). Electronic banking, being one of the greatest innovations in the development of financial services sector, is the mechanism through which financial transactions are carried out electronically without the existence of a physical branch, in accordance with Ombati et al. (2010). However, Beerli et al. (2004) claimed that most banks today offered mainly the same type of products while the main product was usually not the attribute that makes a client loyal.

2.2 Customer Satisfaction and the Service Quality of E-Banking

Customer satisfaction is seen as the way in which customers react to the perceived gap between the anticipated and delivered quality of products (Tse & Wilton, 1988). Likewise, Kotler (2003) formulated customer satisfaction as the degree of fulfillment according to which an individual's expectations of a product's performance were met. Kumar et al. (2009) concluded that exceptional service quality led to satisfied customers and strengthened customer loyalty.

Despite the fact that service quality is one of the identified aspects of judgement contributing to customer satisfaction, it could be considered at many levels and dimensions (Caruana, 2002). Constructed by Crosby (1979), service quality is seen as an elusive and ambiguous concept that could not be easily articulated by customers.

Based on the discussions and analysis of the links between customer satisfaction and perceived expectations of service quality, the SERVQUAL model attempts to operationalize the concept of satisfaction in a theoretical and academic context (Allen & Rao, 2000). SERVQUAL is probably the most common model for measuring perceived customer satisfaction (Bahia & Nantel, 2000) in terms of five dimensions: safety, sensitivity, reliability, responsiveness, and visual impression. Jayawardhena (2004) proposed that questions from these five dimensions of the SERVQUAL model could be used to measure the quality of e-banking services.

2.3 Benefits of Customer Loyalty in the Banking Sector

Dimensions of e-service quality are often considered direct drivers of e-loyalty; a significant number of papers found it positively associated with e-satisfaction and performed a mediator role in its relationship with customer loyalty (Chang & Chen, 2009; Strauss & Frost, 2001). However, various studies focused only on some parts of the relationships between these factors and their impact on e-loyalty (Floh & Treiblmaier, 2006). From another point of view, findings from the study of Atay & Apak (2013) showed that highly educated and more wealthy clients

with inclusive and longer utilization of e-banking services tended to be more satisfied with these services and remained more loyal to the bank.

Although banks have realized that it is much more profitable to retain existing customers than to acquire new ones, there is an increasing tendency among customers to choose multiple financial providers for provision of financial services (Cox & Lasley, 1984). To investigate the importance of customer loyalty to electronic services, identifying the factors that trigger repeated purchasing actions and verbal recommendations is an area that needs to be researched and emphasized (Srinivasan et al., 2002). This is especially true for industries that are highly dependent on their long-term reputation, such as the banking sector. Hence, developing an inclusive approach to obtaining information on the loyalty drivers, whether business-related (Bhattacharjee, 2001) or other individual attributes (Mägi, 2003), provides opportunities for banks to effectively leverage the value derived from the loyal customer base.

2.4 E-Banking and Bank Performance

Studies assessing the performance of new financial services from the lucrative perspective of developing these services, dated back much earlier than the beginning of the exponential growth trend in the use of e-banking. While the interest from lending activities is the main source of profit for most commercial banks, they are not considered to be sufficient to economically sustain their financial stability, so sources of non-interest income are required (Chen et al., 2017). Nowadays, non-interest income deriving mainly from innovative finance enables banks to strategically diversify income sources as presented in their financial statements (Beck et al., 2016). Even from the cost perspective, incomes from e-banking services impact the profits of commercial banks (Attafuah et al., 2024). Cost issues could indeed deter many technological innovations in the financial sector (Scarbrough & Lannon, 1989).

For banks offering retail services, cross-selling is the backbone of their marketing approach (Meidan, 1984). Storey & Easingwood (1999) demonstrated that in consumer financial services, the benefit of innovative products was reflected in more than one dimension and they highlighted the critical determinants of success in the context of financial services. While sales and profits were important, strategic impact at the company level contributed about half of the total value. They argued that true performers offered multiple benefits, supported by internal alignment to strong execution of marketing strategies, to complement existing offerings, strengthen brand image, diversify the service portfolio, and prepare the organization for future growth. These strategic factors explain why innovative financial products could still generate long-term value despite unfavorable financial indicators. Another benefit presented by the authors was a better understanding of the specific requirements of a market, to facilitate easier identification of further opportunities.

2.5 Theoretical Framework of Consumer Price Behavior

By introducing heuristic price theory, Thomas (2023) suggested that price decisions are multidimensional and each of them uses several rules influenced by both conscious and unconscious reactions. Main types of responses guiding the rules include, comparing prices, linking price and quality, judging fairness, weighting price against features, price negotiating option and the pain of paying. To understand how prices influence consumer choices under certain context, it is important to establish which decision rules being applied.

Therefore, in addition to the traditional dimensions of service quality, this study included the price of e-banking services to test whether it had a significant impact on customer loyalty in the banking sector of developing countries.

2.6 E-Banking Services in the Kosovo Banking Sector

The functioning of the banking sector in Kosovo has been fully restored since the 1999 war. “The establishment of the Banking and Payments Authority of Kosovo was created by the international organization, United Nations Interim Administration Mission in Kosovo, in 1999 to oversee the implementation of the monetary and financial framework, to promote an efficient and secure payments system, and to support the creation of a sound financial sector within the territory of Kosovo” (Svetchine, 2005). This organization is acting today as Central Bank of the Republic of Kosovo.

This sector was based on a prudent regulatory and supervisory framework established in accordance with international standards and best practices and now represents one of the main pillars of a market-oriented economy. Developments in this sector in the following years, compared to other transition economies and especially those in the region, have contributed to improving public confidence in the banking system and strengthening the intermediary function of banks in the Kosovo economy (Toçi & Tyrbedari, 2005). Foreign capital continues to dominate most sectors of the financial system, with domestically-owned banks accounting for only 15.7% in December 2023 (Central Bank of the Republic of Kosovo, 2024).

The assets of the Kosovo financial system amounting to EUR 12.75 billion in 2024 had increased by 14.8% compared to 2023; they were dominated mainly by banking assets of EUR 8.6 billion and constituted 67% of the assets in the financial sector. The financing of activities in this sector was supported almost entirely by customer deposits and equity, where deposits amounting to EUR 6.92 billion constituted 80.9% of the total liabilities and equity of the sector in 2024. The banking sector continues to be stable, liquid, and well-capitalized with a loan-to-deposit ratio of 84% and a non-performing loan rate at a historically low level of 1.9%. This indicates a high public confidence in the banking system and a significant support to the real economy (Central Bank of the Republic of Kosovo, 2025).

Table 1. Network of the banking sector in the Republic of Kosovo

Description	2019	2020	2021	2022	2023
Number of bank branches	47	50	43	49	50
Number of bank sub-branches	159	149	145	141	152
Number of automated teller machines	497	513	516	534	583
Number of points of service	13,769	13,421	13,836	14,769	17,187
Number of e-banking accounts	337,693	411,346	537,733	688,891	851,645

Note: Source: Central Bank of the Republic of Kosovo Financial Stability Report, No. 20 (Central Bank of the Republic of Kosovo, 2024).

The data in Table 1 depicts the evolution of the banking network in Kosovo by depicting the gradual transition from physical banking services (bank branches and sub-branches) toward digital services (automated teller machines, points of services and e-banking accounts) of the banking sector in Republic of Kosovo from 2019 to 2023. The number of bank branches remains relatively stable throughout this period, with 50 branches in 2023 compared to 47 in 2019. The number of sub-branches shows overall downward trend for the period 2019 to 2022 despite a small increase from 2022, resulting in 152 sub-branches in 2023. While, a consistent growth each year is shown in the number of automated teller machines through this period, with 583 operating ATMs in the country by the end of 2023, from 497 in 2019. A strong expansion of number of point of services is marked with strong increase especially after 2021, with 17,187 points of sales in 2023 in comparison with 13,769 in 2019. The number of e-banking accounts, through which online banking services were performed, reached 851,645 at the end of 2023, representing an annual growth of 23.6%.

The number of payments performed through e-banking reached over 9.7 million transactions, representing 17% increase compared to the previous year, with a value of over 21.4 billion Euros and an annual growth of 13% (Central Bank of the Republic of Kosovo, 2024). At the end of 2024, commercial banks of the Republic of Kosovo operated with a total of 58 branches and 166 sub-branches in the country (Central Bank of the Republic of Kosovo, 2025).

The total number of valid bank accounts reached 2.51 million and the trend of increasing digitalization of payments has continued at a rapid pace in recent years. However, despite these increases, the number of branches and sub-branches remains largely unchanged and does not show a decline as might be assumed as a result of double-digit increases in e-banking accounts and online transactions over the last 5-year period (Central Bank of the Republic of Kosovo, 2025). The data presented on the banking services network of the Republic of Kosovo over the period 2019–2023 signifies consolidation of tradition physical infrastructure along with rapid expansion of digitalized banking services.

3. Methodology

For the collection of primary data, a self-administered questionnaire was designed with Google Form and then distributed electronically to bank clients across the region in the Republic of Kosovo, resulting in the return of 208 valid questionnaires. Participation was voluntary and respondents completed the questionnaires anonymously. Due to time constraint and respondents' accessibility to data, the study applied a convenience sampling approach (Bryman & Bell, 2007) which might narrow the representativeness of the sample. Owing to these limitations, the findings mainly reflect trends among e-banking users but not the views of all bank clients in Kosovo. Therefore, upcoming research could increase the sample size and adopt random sampling to improve generalizability.

The five dimensions of e-banking service such as reliability, responsiveness, security, sensitivity, and tangible objects have been included by adapting the modified version of the e-SERVQUAL conceptual model from Parasuraman et al. (1985). The questionnaire had a total of 31 questions, in which the first part was composed of demographic questions and the second part included questions related to the impact of e-banking service quality, and the last part contained questions related to other factors that might affect customer satisfaction and loyalty to banks.

The research questions and main hypotheses of this study included:

Q: How does the quality of e-banking services affect the satisfaction and loyalty of bank customers in Kosovo?

H₁: Dimensions of e-banking service quality have a positive impact on customer satisfaction in the banks of Kosovo.

H₂: Dimensions of e-service quality have a positive impact on customer loyalty in the banks of Kosovo.

H₃: The price of e-services has a negative impact on customer loyalty in the banks of Kosovo.

H₄: Demographic factors such as age, education, and the level of income have a statistically significant impact on customer loyalty in the banks of Kosovo.

SPSS software was employed in data processing and data analysis was conducted through econometric tests. The statistical analyses and tests in this empirical research involved correlation analysis, regression analysis, factorial analysis, and analysis of variance (ANOVA); the data were tested for multicollinearity using the Variance Inflation Factor (VIF).

Correlation Analysis

Before exploring causal relationships, correlation analysis was conducted to measure the strength and direction of linear relationships between the dimensions of e-banking service quality and client satisfaction (H₁), between e-banking service quality and client loyalty (H₂), as well as between price and client loyalty (H₃).

Regression Analysis

Regression analysis was applied to test how the dimensions of service quality and price affected client satisfaction and loyalty by estimating the magnitude and significance of these relationships in hypotheses H₁, H₂, and H₃.

Factorial Analysis

Considering that e-banking service quality had various dimensions, factorial analysis was applied to identify the main factors and validated the constructs for analysis. It could ensure that the measured variables could accurately reflect the theoretical framework being analyzed.

ANOVA

ANOVA was used to reveal if there were significant differences in the average scores of the loyalty and satisfaction among customer groups receiving different levels of e-banking service quality. This analytical technique was deemed suitable for comparing the means of several groups in one go and determine how changes in the service quality affected clients. This method was used to analyze the impact of demographic factors on client loyalty (H₄).

4. Results

This section is further divided into sub-sections to describe results concisely and precisely, provide interpretations, and draw possible conclusions from the results.

Table 2. Demographic data of respondents

Questions	Frequency	Percentage
Sex	N = 208	100
Female	107	51.4
Male	101	48.6
Age groups	N = 208	100
18-30	45	21.6
31-45	75	36.1
46-60	81	38.9
>60	7	3.4
Education levels	N = 208	100
Secondary School	17	8.2
Bachelor	93	44.7
Master	84	40.4
PhD	14	6.7
Monthly income	N = 208	100
<350	11	5.3
350-1,000	96	46.2
1,500-2,000	65	31.3
>2,000	36	17.3

The data presented in Table 2 provides an overview of the demographic structure of 208 individuals; these include gender, age group, level of education, and monthly income. In terms of gender, the group was almost evenly divided with 51.4% females and 48.6% males. In terms of age groups, the majority of participants were

categorized into the 31–45 and 46–60 age ranges, which accounted for 36.1% and 38.9% in this classification, respectively. The smallest group of only 3.4% included those over age 60, thus indicating a small representation of the elderly.

In terms of education levels, the majority of participants had a Bachelor's degree (44.7%) or a Master's degree (40.4%), indicating a high level of education among the respondents. Only a small percentage had completed secondary school (8.2%) or had a PhD (6.7%).

The majority of participants had a monthly income between 350 and 1,000 Euros (46.2%), followed by those with monthly income between 1,500 and 2,000 Euros (31.3%). Only 5.3% had monthly income of less than 350 Euros while 17.3% had monthly income of over 2,000 Euros.

Table 3. Results of correlation analysis

Variables		(1)	(2)	(3)	(4)	(5)	(6)	(7)
Satisfaction (1)	Cor.	1	0.190**	0.554**	0.482**	0.485**	0.478**	0.374**
	Sig.		0.006	0.000	0.000	0.000	0.000	0.000
	N	208	208	208	208	208	208	204
Loyalty (2)	Cor.	0.190**	1	-0.066	-0.010	-0.085	-0.067	-0.115
	Sig.	0.006		0.344	0.885	0.223	0.335	0.102
	N	208	208	208	208	208	208	204
Reliability (3)	Cor.	0.554**	-0.066	1	0.614**	0.732**	0.605**	0.624**
	Sig.	0.000	0.344		0.000	0.000	0.000	0.000
	N	208	208	208	208	208	208	204
Responsiveness (4)	Cor.	0.482**	-0.010	0.614**	1	0.649**	0.656**	0.534**
	Sig.	0.000	0.885	0.000		0.000	0.000	0.000
	N	208	208	208	208	208	208	204
Safety (5)	Cor.	0.485**	-0.085	0.732**	0.649**	1	0.704**	0.657**
	Sig.	0.000	0.223	0.000	0.000		0.000	0.000
	N	208	208	208	208	208	208	204
Sensitivity (6)	Cor.	0.478**	-0.067	0.605**	0.656**	0.704**	1	0.602**
	Sig.	0.000	0.335	0.000	0.000	0.000		0.000
	N	208	208	208	208	208	208	204
Tangible objects (7)	Cor.	0.374**	-0.115	0.624**	0.534**	0.657**	0.602**	1
	Sig.	0.000	0.102	0.000	0.000	0.000	0.000	
	N	204	204	204	204	204	204	204

Note: **The correlation was significant at the 0.01 (1%) level.

Based on the results of the Table 3, correlation analysis revealed that customer satisfaction had a positive relationship with all the other variables. However, the strongest positive relationship was between satisfaction and the dimension of reliability (coefficient $r = 0.554$), where the coefficient was statistically significant at the 1% significance level ($p < 0.001$). Customer satisfaction with the bank had the weakest positive relationship with the quality variable of tangible objects (coefficient $r = 0.374$), where the coefficient was statistically significant at the 1% significance level ($p < 0.001$).

Table 4. Results of reliability analysis

Variables	Cronbach's Alpha	No. of Questions
Reliability	0.878	3
Responsiveness	0.908	3
Safety	0.904	3
Sensitivity	0.912	3
Tangible objects	0.911	3

Table 4 presents the results of the Cronbach's alpha test, which was used to demonstrate the reliability or consistency of questions related to the quality variables. According to the results presented above, there were three questions in each category of quality and the value of Cronbach's alpha test was very high for all variables, suggesting great consistency in the questionnaire for obtaining valid results.

Table 5 presents the results of the factor analysis, which showed that the first two components were the most important, as they explained a large part of the total variance in the data. The first component, with a value of 3.92 and 56.11% of the variance, was mainly related to the variables, reliability, responsibility, safety, sensitivity, and tangible objects. These results indicated that this component represented a general dimension of reliability and safety, thus highlighting the importance of these aspects in the analyzed data.

The second component, which had a value of 1.12 and explained 16.05% of the variance, represented a different

dimension of the data related mainly to the loyalty and satisfaction variables. This component illustrated that the aspects of loyalty and satisfaction were more important and distinct from the first dimension. The combination of these two components together explained 72.17% of the total variance, as they captured most of the important information in the data.

Table 5. Factorial analysis

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	3.928	56.115	56.115	3.928	56.115	56.115
2	1.124	16.058	72.174	1.124	16.058	72.174
3	0.541	7.722	79.896			
4	0.471	6.733	86.630			
5	0.369	5.267	91.896			
6	0.329	4.700	96.597			
7	0.238	3.403	100.000			

Variables	Component	
	1	2
Loyalty	-0.058	0.941
Satisfaction	0.675	0.441
Reliability	0.853	-0.013
Responsiveness	0.810	0.038
Safety	0.877	-0.088
Sensitivity	0.835	-0.056
Tangible objects	0.786	-0.178

The results of linear regression model showed that the correlation (R) between the independent variables and dependent variable, Satisfaction, was 0.60, suggesting a strongly positive relationship. The R^2 value (0.36) indicated that 36% of the variance in satisfaction could be explained by the model or the independent variables presented in the model.

The ANOVA results illustrated that the regression model was significant ($F = 22.269, p < 0.001$), so the results of coefficients were valid and their interpretation is presented below:

Reliability had an unstandardized coefficient of 0.121 and was statistically significant ($p < 0.001$), with a standardized Beta value of 0.38. This suggests that an increase in perceived reliability was associated with an increase in Satisfaction.

Responsiveness had a coefficient of 0.05 and a small p -value ($p = 0.058$), indicating a positive relationship though not at the statistically significant level.

Security had a low coefficient of 0.013 and was not statistically significant ($p = 0.718$), indicating that perceived Security did not have a significant impact on satisfaction in this model.

Sensitivity had a coefficient of 0.059 and was on the verge of statistical significance ($p = 0.049$), suggesting a small but significant impact on satisfaction.

Tangible objects had a negative coefficient of -0.025 and was not statistically significant ($p = 0.363$), indicating that this factor did not have a significant impact on Satisfaction in this model.

Table 6 presents the results that showed that reliability was the main factor influencing customer satisfaction, being the only variable that had a significant and large positive impact. Responsiveness and sensitivity also had positive impacts, but smaller and less statistically significant. meanwhile, security and tangible objects did not have significant impacts on satisfaction.

In this linear regression model presented in Table 7, the correlation (R) between the independent variables and the dependent variable loyalty was 0.138, indicating a weakly positive relationship. The R^2 value (0.019) indicated that only 1.9% of the variance in loyalty could be explained by the independent variables in the model.

The results of the ANOVA showed that the regression model was not significant ($F = 0.774, p = 0.570$), meaning that the independent variables together did not explain the differences in loyalty.

Considering these results, the overall model was not statistically significant, therefore the interpretation of individual coefficients should be made with caution.

Reliability had an unstandardized coefficient of 0.001 and was not statistically significant ($p = 0.969$), indicating that perceived trustworthiness did not have a significant impact on loyalty.

Responsiveness had a coefficient of 0.028 and was not statistically significant ($p = 0.300$), indicating a weakly and insignificantly positive relationship.

Security had a negative coefficient of -0.018 and was not statistically significant ($p = 0.601$), suggesting that perceived security did not significantly affect loyalty.

Sensitivity had a negative coefficient of -0.009 and was not statistically significant ($p = 0.773$), indicating that sensitivity did not have a significant impact on loyalty.

Table 6. Results of the first model–satisfaction

Summary of the Model						
Model	<i>R</i>	<i>R</i> ²	Adjusted <i>R</i> ²		Std. Error of the Estimate	
1	0.600 ^a	0.360	0.344		0.68640	
ANOVA ^a						
Model		Sum of Squares	<i>df</i>	Mean Square	<i>F</i>	Sig.
1	Regression	52.460	5	10.492	22.269	0.000
	Residual	93.286	198	0.471		
	Total	145.745	203			
Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	<i>t</i>	Sig.
		<i>B</i>	Std. Error	Beta		
1	(Constant)	2.828	0.258		10.955	0.000
	Reliability	0.121	0.028	0.380	4.290	0.000
	Responsiveness	0.050	0.026	0.156	1.907	0.058
	Safety	0.013	0.035	0.036	0.362	0.718
	Sensitivity	0.059	0.030	0.172	1.979	0.049
	Tangible objects	-0.025	0.028	-0.074	-0.911	0.363

Note: a. Dependent variable: Satisfaction

Table 7. Results of the second model–loyalty

Model Summary						
Model	<i>R</i>	<i>R</i> ²	Adjusted <i>R</i> ²		Std. Error of the Estimate	
1	0.138 ^a	0.019	-0.006		0.69731	
ANOVA ^a						
Model		Sum of Squares	<i>Df</i>	Mean Square	<i>F</i>	Sig.
1	Regression	1.881	5	0.376	0.774	0.570
	Residual	96.275	198	0.486		
	Total	98.157	203			
Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	<i>t</i>	Sig.
		<i>B</i>	Std. Error	Beta		
1	(Constant)	4.157	0.262		15.854	0.000
	Reliability	0.001	0.029	0.004	0.039	0.969
	Responsiveness	0.028	0.027	0.105	1.039	0.300
	Safety	-0.018	0.035	-0.064	-0.524	0.601
	Sensitivity	-0.009	0.030	-0.031	-0.289	0.773
	Tangible objects	-0.032	0.028	-0.113	-1.126	0.262
	Price	-0.381	0.052	0.274	-7.32	0.000

Note: a. Dependent variable: Loyalty

Tangible objects had a negative coefficient of -0.032 and was not statistically significant ($p = 0.262$), indicating that tangible objects did not affect loyalty.

The linear regression model did not significantly explain the variance in loyalty and there were no independent variables that were statistically significant in this model. This indicates that the factors selected for this analysis did not sufficiently explain and predict customer loyalty.

The results of the regression analysis showed that changes in the values of the price variable were related to changes in the values of the loyalty variable. The unstandardized coefficient (*B*) for the price is -0.381, indicating that, on average, for each unit increase in price, the variable loyalty was expected to decrease by 0.381 units. These results indicate a negative relationship between price and loyalty, suggesting that higher prices were associated with lower levels of loyalty. Furthermore, the *t*-test for the price coefficient was statistically significant ($t = 7.32$, $p < 0.001$), meaning that this relationship was statistically significant.

Table 8 shows the result of the VIF test, for testing variables for multicollinearity. According to the data presented, all independent variables had a VIF test value lower than 5, indicating that the independent variables did not have a strong correlation with each other, so the problem of multicollinearity was not present.

Table 8. Testing data for multicollinearity

Variables	Collinearity Statistics	
	Tolerance	VIF
Reliability	0.412	2.427
Responsiveness	0.481	2.077
Safety	0.332	3.01
Sensitivity	0.425	2.35
Tangible objects	0.494	2.025

Note: VIF = Variance Inflation Factor.

Table 9 offers the results of the ANOVA to show statistically significant differences in loyalty according to social and demographic data.

Table 9. Results of analysis of variance ANOVA analysis regarding the impact of demographic factors on bank loyalty

Variables		Sum of Squares	df	Mean Square	F	Sig.
Gender	Between groups	0.720	3	0.240	0.955	0.415
	Within groups	51.237	204	0.251		
	Total	51.957	207			
Age	Between groups	7.648	3	2.549	3.871	0.010
	Within groups	134.333	204	0.658		
	Total	141.981	207			
Education	Between groups	5.888	3	1.963	3.717	0.012
	Within groups	107.722	204	0.528		
	Total	113.611	207			
Monthly income	Between groups	5.276	3	1.759	2.592	0.054
	Within groups	138.397	204	0.678		
	Total	143.673	207			
Residence	Between groups	0.132	3	0.044	0.578	0.630
	Within groups	15.479	204	0.076		
	Total	15.611	207			

The results of ANOVA for the gender variable showed that the F -value was 0.955 with a p -value of 0.415. Since the p -value was greater than the usual alpha level of 0.05, we concluded that there were no statistically significant differences in the means of the dependent variable in different gender groups. Therefore, gender did not have a statistically significant impact on loyalty.

For the age group variable, the F -value was 3.871 with a p -value of 0.010. Since the p -value was less than the usual alpha level of 0.05, we concluded that there were statistically significant differences in the means of the dependent variable according to the age of the respondents. Therefore, age group had a statistically significant impact on loyalty.

As regards the education variable, the F -value was 3.717 with a p -value of 0.012. Since the p -value was less than the usual alpha level of 0.05, we concluded that there were statistically significant differences in the means of the dependent variable at different educational levels. So, education had a statistically significant impact on loyalty.

Regarding monthly income, the F -value was 2.592 with a p -value of 0.054. Since the p -value was below the alpha level of 0.1, the differences in the means of the dependent variable across different income group were statistically significant at the 10% level ($p < 0.1$). So, income had a significant impact on customer loyalty.

The residence variable had an F -value of 0.578 with a p -value of 0.630. Since the p -value was much larger than the usual alpha level of 0.05, there were no statistically significant differences in the means of the dependent variable across different residence groups. So, residence had no statistically significant impact on customer loyalty.

The ANOVA test revealed that age group, education, and income level had a significant impact on customer loyalty, with p -values < 0.05 . Meanwhile, gender did not show a significant impact on loyalty towards banks in Kosovo.

5. Discussion

Hypothesis testing

H₁: Dimensions of e-service quality have a positive impact on customer satisfaction in the banking sector of the Republic of Kosovo—Partially accepted

The results of the analysis illustrated that dimensions of e-service quality had different impacts on customer satisfaction in Kosovo banks. With a coefficient of 0.121 and a very low p -value ($p < 0.001$), reliability turned out to be the main factor affecting customer satisfaction, with a Beta value of 0.380. In addition, responsiveness had a coefficient of 0.050 and a p -value within the 10% statistical significance level ($p < 0.1$), while sensitivity had a coefficient of 0.059 and a p -value of 0.049, suggesting a small but significant impact on satisfaction. However, safety had a low coefficient of 0.013 and a high p -value ($p = 0.718$), while tangible objects had a negative coefficient of -0.025 and a p -value of 0.363, indicating that these factors did not have significant impacts on satisfaction.

Based on these results, hypothesis H_1 could be partially accepted, considering that some of the dimensions of e-service quality such as reliability, responsiveness, and sensitivity had a positive impact on customer satisfaction. However, security and tangible objects did not have a significant impact.

H₂: Dimensions of e-service quality have a positive impact on customer loyalty in Kosovo banks—Rejected

The analysis of the linear regression model showed that the model failed to significantly explain the variation in loyalty and there were no independent variables that were statistically significant in this model. The interpretation of individual coefficients should be made with caution. One possible explanation is that loyalty may be influenced by other factors not included in the model.

The results of the analysis illustrated that the dimensions of e-service quality did not have a statistically significant impact on customer loyalty in Kosovo banks, as shown by the coefficients and p -values for each of them.

Perceived reliability did not show a significant impact on loyalty, having a coefficient of 0.001 and a very high p -value of 0.969). Responsiveness had a coefficient of 0.028 and a p -value of 0.300, suggesting a weak and insignificant positive relationship. Security had a negative coefficient of -0.018 and a p -value of 0.601, indicating that perceived security did not affect loyalty. Sensitivity had a negative coefficient of -0.009 and a p -value of 0.773, while tangibles had a negative coefficient of -0.032 and a p -value of 0.262, indicating that sensitivity and tangible objects did not have a significant impact on loyalty.

Based on these results, hypothesis H_2 was rejected as none of the dimensions of electronic service quality had shown a positive and significant impact on customer loyalty among the bank customers in the Republic of Kosovo.

H₃: The price of electronic services has a negative impact on customer loyalty towards banks in Kosovo—Accepted

Based on the results of the regression analysis, we could test hypothesis H_3 using the coefficient determined for the price variable. To test this hypothesis, the value of the coefficient determined for price, $B = -0.381$, was taken as a basis, together with the t -test value for this coefficient, $t = 7.32$, with a significance level of $p < 0.001$.

The coefficient determined for Price was negative (-0.381), indicating a negative relationship between price and loyalty. The t -test for the price coefficient was statistically significant ($t = 7.32$, $p < 0.001$).

Given these regression results, hypothesis H_3 was supported because the price coefficient was negative and statistically significant, indicating a negative relationship between the price of electronic services and customer loyalty to the banks in Kosovo.

H₄: Age, education, and monthly income of customers have a statistically significant impact on loyalty to the banks in Kosovo—Accepted

The results of ANOVA for hypothesis H_4 indicated that socio-demographic factors had a statistically significant impact on loyalty to banks in Kosovo. The gender analysis showed that the F -value was 0.955 with a p -value of 0.415, which was higher than the usual significance level of 0.05, indicating that there were no statistically significant differences in the means of the dependent variable in different gender groups.

On the other hand, the analysis of age group, education, and income levels demonstrated statistically significant differences in the means of the dependent variables in the different groups. For age group, the F -value was 3.871 with a p -value of 0.010, for education the F -value was 3.717 with a p -value of 0.012, while for income levels the F -value was 2.592 with a p -value of 0.054. These results revealed that age group, education, and income levels had a statistically significant impact on loyalty at different levels of significance. Regarding the residence variable, the F -value was 0.578 with a p -value of 0.630, which was greater than the usual alpha level of 0.05, suggesting that there were no statistically significant differences in the means of the dependent variable in different residence groups. Thus, residence did not have a statistically significant impact on loyalty.

Based on these results, it appears that socio-demographic factors such as age group, education and income levels have a significant impact on loyalty to banks in Kosovo, while gender and residence do not have a significant impact. Hypothesis H_4 could be accepted. This is because demographic factors such as age group, education, and monthly income exhibit a statistically significant impact on loyalty toward banks of the Republic of Kosovo.

6. Conclusions and Recommendations

From the findings of this research, it is concluded that reliability as a dimension of e-banking service quality turned out to be the main factor that positively influenced customer satisfaction in Kosovo banks. The second

significant factor was responsiveness, followed by sensitivity, with a positive impact on customer satisfaction. Since none of the dimensions of e-service quality had shown a positive and significant direct impact on customer loyalty to the banks in Kosovo, other factors that might have an impact on their loyalty should be identified. According to the results of the analysis, the negative relationship between price and loyalty of bank customers in Kosovo suggested that banks should review the prices and commissions of e-banking services to retain their customers. The demographic factors that resulted in an impact on the loyalty of bank customers in Kosovo were age, education and monthly income levels which represented the most common customer segmentation by commercial banks when defining products and services as well as their pricing and tariff strategies.

There could be several reasons why e-banking service quality did not have a strong effect on loyalty in this study. Although clients might perceive e-banking services as reliable and responsive, this does not always mean it leads to greater loyalty. Consumers often compare fees and other benefits, so they may not stay loyal if another bank offers similar e-banking quality, with more reasonable prices for them. Besides this, the link between e-banking service quality and client loyalty can vary in markets where digital skills differ across demographic groups. Younger or more digitally educated clients may value better e-banking service quality, while older or less digitally skilled clients might still prefer traditional banking. This makes e-banking service quality alone an insufficient explanation for loyalty.

Overall, these findings showed that in the banking sector of the developing countries like Kosovo, e-banking service quality by itself might not be sufficient to drive client loyalty though price and demographic categories were important determinants. It appears that a broader integration of other factors such as past experience, trust, and bank reputation should be considered. These regression results offered preliminary results as indicative trends that required additional research with a larger and more diverse sample and different predictors to be included.

6.1 Practical Implications for Commercial Banks

Considering that customer loyalty is impacted by how services are priced and who the customers are, the management of the commercial banks should maintain transparent and competitive pricing models. Customers are more likely to stay loyal if they perceive that e-banking services are fairly-priced. Excessive fees or hidden charges can erode trust and reduce loyalty. Therefore, banks are recommended to tailor pricing strategies by using differentiated loyalty programs according to customer's age, education, and income levels to incentivize long-term client engagement with their e-banking services. In addition, commercial banks in Kosovo should identify the quality factors considered to be important to their customers by noting which of these factors had an effect on the perception of e-banking service quality, hence affecting customer satisfaction, loyalty to the bank, and consequently their lasting competitive advantages.

6.2 Practical Implications for Central Banks and Financial Regulators

The findings of the research offered valuable insights for institutions responsible to oversee service quality, pricing adequacy, and demographic inclusion which impacted both financial system stability and customers' protection. Unfair or non-transparent e-banking charges may trigger liquidity issues at commercial bank level, especially in the banking sector highly dependent on customers' deposits. Therefore, regulatory frameworks should encourage fairly-priced online banking services to strengthen trust in banks, in order to mitigate funding risk during periods of financial distress. Moreover, central banks are recommended to monitor whether online banking pricing models create barriers to financial inclusion, especially for disproportionately affected demographic groups within developing and emerging economies to ensure that these services are accessible and affordable to all segments of the population. Evidently, e-banking service as a financial innovation has transformed banking operations by enhancing access, efficiency, service delivery, and transparency. However, it has introduced new risks in the financial system, such as cyber risk. Therefore, a balanced approach that encourages innovation while strengthening consumer protection is critical for regulators to ensure positive impact on e-banking in respect of financial stability of the developing and emerging countries.

6.3 Future Directions

The limitation of this study is the non-inclusion of rural customers in the empirical analysis, as their representation in the sample was almost non-existent. This would have enabled a comprehensive result in terms of understanding perceptions of e-banking service quality and loyalty; as for rural customers, considering e-banking enabled remote access to bank accounts.

Future research should integrate other contextual variables and examine the interaction of other factors such as cultural norms, operational quality, and perceived risks in shaping perceptions of e-banking service quality, satisfaction, and loyalty of bank customers in developing and emerging economies.

Author Contributions

Conceptualization, M.M., D.K., and F.M.; methodology, F.M.; software, D.K.; validation, M.M., D.K., and F.M.; formal analysis, M.M.; investigation, M.M.; resources, M.M.; data curation, F.M.; writing—original draft preparation, M.M.; writing—review and editing, M.M.; visualization, M.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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